

- AES-128

AES-128+SPECK

AES-256

AES-256+AES-128

AES-256+ChaCha20 (AES-NI on)

AES-256+ChaCha20 (ChaCha-SIMD on)

AES-256+ChaCha20 (both off)

AES-256+ChaCha20+AES-128 (AES-NI on)

AES-256+ChaCha20+AES-128 (ChaCha-SIMD on)

AES-256+ChaCha20+AES-128 (both off)

ChaCha20

ChaCha20+SPECK

RECTANGLE
- ECC handshake share of total time — selected entities (no/partial hardware acceleration)
- The graph displays the percentage share of total time for ECC handshake across different data sizes (1KB to 50MB) for various cryptographic configurations. The y-axis is logarithmic, ranging from 0.0 to 100.0. The x-axis represents data size. Most configurations show a decreasing trend in the ECC share as data size increases. ChaCha20+SPECK and ChaCha20 generally show higher ECC shares compared to other configurations, especially at larger data sizes.
- | | 1KB | 5KB | 10KB | 50KB | 100KB | 1MB | 5MB | 10MB | 50MB |
|---|------|------|------|------|-------|-----|-----|------|------|
| AES-128 | 86.2 | 58.8 | 42.2 | 12.9 | 6.9 | 0.7 | 0.1 | 0.1 | 0.0 |
| AES-128+SPECK | 84.8 | 55.9 | 39.3 | 11.5 | 6.1 | 0.6 | 0.1 | 0.1 | 0.0 |
| AES-256 | 82.4 | 51.0 | 34.5 | 9.7 | 5.1 | 0.5 | 0.1 | 0.1 | 0.0 |
| AES-256+AES-128 | 73.2 | 38.4 | 24.0 | 6.0 | 3.1 | 0.3 | 0.1 | 0.0 | 0.0 |
| AES-256+ChaCha20 (AES-NI on) | 93.5 | 84.0 | 74.6 | 39.3 | 24.7 | 3.1 | 0.6 | 0.3 | 0.1 |
| AES-256+ChaCha20 (ChaCha-SIMD on) | 88.0 | 66.9 | 51.5 | 18.1 | 10.0 | 1.1 | 0.2 | 0.1 | 0.0 |
| AES-256+ChaCha20 (both off) | 79.9 | 48.7 | 32.7 | 9.0 | 4.7 | 0.5 | 0.1 | 0.0 | 0.0 |
| AES-256+ChaCha20+AES-128 (AES-NI on) | 91.6 | 80.8 | 70.4 | 34.8 | 21.3 | 2.6 | 0.5 | 0.3 | 0.1 |
| AES-256+ChaCha20+AES-128 (ChaCha-SIMD on) | 82.4 | 56.0 | 40.0 | 12.2 | 6.5 | 0.7 | 0.1 | 0.1 | 0.0 |
| AES-256+ChaCha20+AES-128 (both off) | 71.4 | 37.3 | 23.5 | 5.9 | 3.0 | 0.3 | 0.1 | 0.0 | 0.0 |
| ChaCha20 | 95.8 | 88.7 | 81.4 | 48.9 | 32.6 | 4.5 | 0.9 | 0.5 | 0.1 |
| ChaCha20+SPECK | 93.7 | 80.7 | 69.0 | 31.6 | 18.9 | 2.2 | 0.4 | 0.2 | 0.0 |
| RECTANGLE | 84.6 | 56.6 | 38.6 | 11.0 | 5.9 | 0.6 | 0.1 | 0.1 | 0.0 |